A Supplementary Materials

A.1 Examples of Training Materials

Figures 1 and 2 show examples from the standard RE training materials, including instructional slides and the practice scenario.

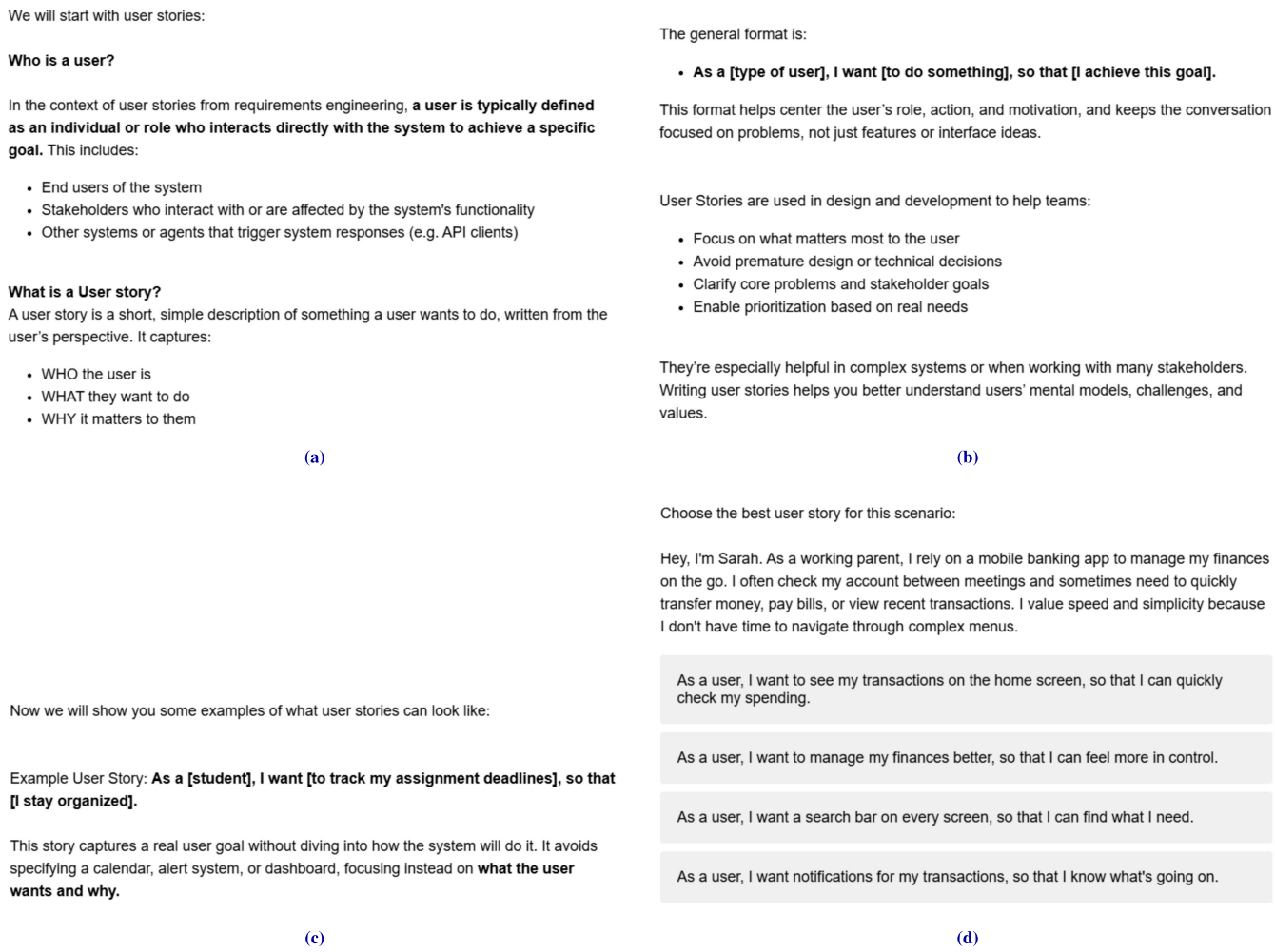


Figure 1: Example instructional materials from the RE training condition.

Your task is to generate user stories from the previous scenario. Consider these pre-writing prompts before you begin:

Pre-writing prompts: As you write user stories, think about:

- Who might be harmed or excluded if this system is poorly designed?
- What kind of resources (time, data, workforce, etc.) do the stakeholders have?
- What are really the main issues the stakeholders are facing?
- What is the stakeholder's ultimate goal?

Start by identifying distinct users and their core needs. For this example you might identify:

- Hospital administrator
- Doctors in training
- Patients
- IT or compliance officer

(a)

Then follow the basic user story template:

- As a [type of user], I want [to do something], so that [I achieve this goal].

For example, one user story you might come up with for the hospital administrator might be:

- As a [hospital administrator] , I want [to continue training our doctors with high quality training materials], so that [we can continue to ensure quality patient care].

Other example user stories for this scenario:

- As a [doctor in training], I want [to access MRI cases], so that [I can practice diagnosis].
- As an [IT or compliance officer], I want to [audit logs], so that [we can ensure HIPAA compliance and patient data safety].

(b)

Come up with your own user stories for each stakeholder that you identify as relevant in the scenario. There is no limit to the number of user stories you can write. Place one user story per line, and follow the template of "As a type of user, I want to do something, so that I achieve this goal."

There is no right or wrong. Try your best to **identify the key issues that the stakeholders are facing and what type of resources they have.**

(c)

Figure 2: Example practice materials from the RE training condition.

A.2 Rubric Coding Examples

Table 13 provides example user stories scored using the Negotiable–Valuable–Testable (NVT) rubric. Table 14 provides example acceptance criteria scored using the Verifiable–Atomic–Implementation-Independent (VAI) rubric.

Table 13: Example user stories evaluated using the Negotiable–Valuable–Testable (NVT) rubric.

User Story	N	V	T	Rationale
As an applicant, I want to receive confirmation when my application is submitted, so that I know it was received successfully.	1	1	1	States a desired outcome without prescribing a specific implementation; clearly benefits the applicant by reducing uncertainty about submission status; receipt of confirmation can be objectively verified.
As a school administrator, I want to reach as many students as possible, so that counselors have accurate data on how to help the entire student body.	1	0	0	Expresses a broad goal without locking in a solution; the benefit to the administrator is indirect and underspecified; success is not defined in measurable or testable terms.
As a school counselor, I want to understand which issues are the biggest factors driving poor academic performance, so that I can intervene in helpful ways.	1	0	1	Describes an exploratory goal rather than a specific system behavior; the benefit is framed abstractly and does not specify a concrete stakeholder outcome; identifying and ranking contributing factors can plausibly be evaluated through analysis outputs.
As a worker for the local government, I want to support local people experiencing housing issues, so that more people feel supported and safe.	1	1	0	Articulates a high-level objective without constraining implementation choices; clearly expresses a social benefit for affected community members; outcomes such as feeling supported or safe are subjective and not readily testable in this context.
As a school counselor, I want to see both academic data and personal challenges for each student, so that I get a full picture of their situation.	0	0	1	Specifies required categories of information, constraining design flexibility; the benefit is generic and does not articulate a specific improvement or outcome for the stakeholder; the inclusion of both data types can be directly verified.
As an energy provider company, I want to automate energy distribution, so that all customers receive appropriate power.	0	0	0	Locks in a particular solution approach, limiting flexibility; the benefit is broadly stated and not tied to a specific stakeholder need or context; the story lacks clear criteria for evaluating what constitutes appropriate power delivery.

Note. N = Negotiable; V = Valuable; T = Testable.

Table 14: Example acceptance criteria evaluated using the Verifiable–Atomic–Implementation-Independent (VAI) rubric.

Corresponding User Story	Acceptance Criterion	V	A	I-I
As an engineer, I want to integrate predictive analytics into the system, so that we can forecast power demands based on weather data and optimize grid performance.	Given that weather data is collected and time-stamped every minute, when I apply predictive models, then I must be able to generate forecasts of power demand based on weather patterns.	1	1	1
As a vulnerable resident, I want the system to prioritise my home during an outage or extreme weather, so that I can receive important life-saving help.	Given the infrastructure data is uploaded, I must be able to examine both the new and legacy system details in one view when I access the system dashboard.	1	1	0
As an engineer, I want to develop a system that can dynamically allocate power to areas based on demand, so that we can improve power distribution and prevent outages.	Given that power load data is updated every minute, when I access the power allocation system, then I must be able to view and adjust power distribution dynamically based on demand from different areas.	1	0	1
As a member of the leadership team, I want to make sure that a smart dispatch system is able to prioritize critical calls correctly, so that we can prevent delays in responding to life-threatening emergencies.	Given that calls of varying urgency are coming in, when the system categorizes the call, then it must accurately prioritize life-threatening emergencies over routine ones and ensure no critical call is buried under non-urgent ones.	0	0	1
As a member of the leadership team, I want to evaluate the costs and benefits of implementing cloud-based solutions so that we can make an informed decision within our budget constraints.	Given that the department is constrained by budget, when evaluating potential systems, then the solution must utilize cloud-based services and stay within the allocated budget for technology.	0	0	0

Note. V = Verifiable; A = Atomic; I-I = Implementation-Independent.

A.3 Self-Reported AI Experience

Tables 15 and 16 summarize participants' self-reported theoretical AI knowledge and practical AI development experience, respectively, along with average AI design task performance.

Table 15: Self-reported theoretical AI knowledge and average AI design task performance.

Knowledge Level	Definition	Total (n)	Prolific	Other	Avg. Perf.*
Basic familiarity	I have heard of some of these concepts or read brief overviews, but could not explain them in detail.	12	3	9	11.08
Moderate understanding	I have learned about these topics in a course or from readings and could describe their purpose and general operation.	22	8	14	12.32
Strong understanding	I can explain most of these concepts in depth and understand when and why they are used.	13	9	4	11.85
Expert-level knowledge	I have a deep grasp of these concepts and could teach them or critique their use in technical or research contexts.	2	2	0	14.00

* Average performance reflects mean scores across all four stakeholder scenarios, with a maximum total score of 20.

Table 16: Self-reported practical AI development experience and average AI design task performance.

Experience Level	Definition	Total (n)	Prolific	Other	Avg. Perf.*
No experience	I have never implemented or applied AI methods.	4	1	3	10.75
Minimal experience	I've followed a tutorial or made small modifications to existing AI code, but not implemented anything from scratch.	11	4	7	11.36
Some experience	I've implemented AI techniques in coursework or personal projects, but not in complex or real-world scenarios.	23	9	14	12.65
Substantial experience	I've applied AI methods in complex projects, research, or work settings, and understand how to tune and troubleshoot them.	8	6	2	10.50
Expert-level experience	I have led AI development efforts, published implementation work, or significantly innovated on standard techniques.	3	2	1	14.33

* Average performance reflects mean scores across all four stakeholder scenarios, with a maximum total score of 20.